

Assignment 4 - common pitfalls

- A subsequence may skip characters; a substring is contiguous. LCS is not longest common substring.
- Define DP states, bases and computation order before writing loops. Reconstruct a feasible optimum, not just a number.
- Merge only current adjacent intervals. A plan must be chronological and charge each combined size.
- Tree take/skip uses child states. Empty selection is valid, including all-negative weights.
- Include overlaps. An empty single pattern matches every text boundary; alarm dictionary patterns are nonempty.
- Equal hashes do not prove equal strings. With base 31 and modulus 101, ab and dj collide; verify characters.
- Do not materialize every suffix string. LCP compares the previous SA neighbour; $\text{lcp}[0]=0$.
- Aho-Corasick must count suffix outputs, preserve duplicate dictionary positions, and reset per-text counters.